

Zhengyang Yu

Curriculum Vitae

San Diego & Bay Area

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Education

2025 – Jun 2028 **Bachelor of Science in Computer Science**, *University of California, San Diego*, San Diego, CA

Expected Graduation: June 2028; GPA: 4.0/4.0

Relevant Coursework: Advanced Data Structures, Design and Analysis of Algorithms, Systems Programming, Computer Organization, Machine Learning

2023 – 2025 **Associate of Science in Computer Science**, *Laney College*, Oakland, CA

Projects

2026 **Quotaquarium**, *Personal Project*

- Designed and built a networked C++ product on ESP32-S3 that authenticates multiple accounts, retrieves live data over HTTPS, and presents it through an animated touch-controlled interface with configurable alerts.
- Implemented concurrent FreeRTOS workers, adaptive and rate-limit-aware polling, and PSRAM-backed rendering; added structured logging and runtime diagnostics to improve performance and reliability.

2025 **HTML to ePub Converter**, *Personal Project*

- Engineered a Python content-processing pipeline with modular HTML parsing, metadata embedding, and structured error handling for reliable ePub generation.

2024 **Home Assistant Presence Plugin**, *Personal Project*

- Built a Python integration that converts entry and exit sensor events into real-time room-occupancy state.
- Designed an event-driven REST API workflow with explicit state transitions for reliable smart-lighting automation and easier testing and debugging.

Skills

Programming Python, Java, C++, C

Systems Linux, HTTP/HTTPS, REST APIs, authenticated clients, concurrent and event-driven systems, networking fundamentals

Engineering Data Structures, Algorithms, Object-Oriented Design, Software Testing, Performance Optimization, Debugging

Tools Git, Docker, FreeRTOS, PlatformIO, VS Code, HTML processing, structured logging and runtime diagnostics

AI Tooling Codex-assisted implementation, prompt iteration, refactoring, debugging, code validation, and documentation

Activities

2024–2025 Computer Consultant, Oakland Public Library - Elmhurst Branch

Diagnosed computer and software issues and translated technical solutions into clear guidance for community members.

2025–Present Member, UCSD Envision Makerspace

Prototype embedded and electronics projects using makerspace fabrication tools.